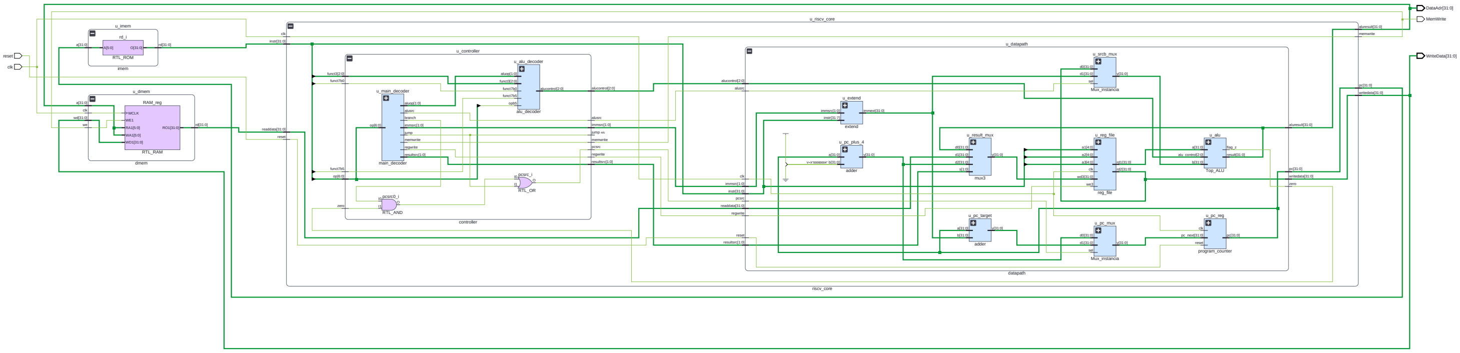
reset
clk
u_imem
rd_i
a[31:0]
A[5:0]
O[31:0]
rd[31:0]
RTL_ROM
imem
u_dmem
RAM_reg
a[31:0]
clk
wd[31:0]
we
WCLK
WE1
RA1[5:0]
WA1[5:0]
WD1[31:0]
RO1[31:0]
rd[31:0]
RTL_RAM
dmem
u_riscv_core
clk
instr[31:0]
readdata[31:0]
reset
u_controller
funct3[2:0]
funct7b0
funct7b5
op[6:0]
zero
u_main_decoder
op[6:0]
aluop[1:0]
alusrc
branch
immsrc[1:0]
jump
memwrite
regwrite
resultsrc[1:0]
main_decoder
u_alu_decoder
aluop[1:0]
funct3[2:0]
funct7b0
funct7b5
opb5
alucontrol[2:0]
alu_decoder
pcsrc0_i
I0
I1
O
RTL_AND
pcsrc_i
I0
I1
O
RTL_OR
alucontrol[2:0]
alusrc
immsrc[1:0]
jump n/c
memwrite
pcsrc
regwrite
resultsrc[1:0]
controller
u_datapath
alucontrol[2:0]
alusrc
clk
immsrc[1:0]
instr[31:0]
pcsrc
readdata[31:0]
regwrite
reset
resultsrc[1:0]
u_extend
immsrc[1:0]
instr[31:7]
immext[31:0]
extend
u_pc_plus_4
a[31:0]
V=X"00000004" b[31:0]
y[31:0]
adder
u_result_mux
d0[31:0]
d1[31:0]
d2[31:0]
s[1:0]
y[31:0]
mux3
u_pc_target
a[31:0]
b[31:0]
y[31:0]
adder
u_srcb_mux
d0[31:0]
d1[31:0]
sel
y[31:0]
Mux_instancia
u_reg_file
a1[4:0]
a2[4:0]
a3[4:0]
clk
wd3[31:0]
we3
rd1[31:0]
rd2[31:0]
reg_file
u_pc_mux
d0[31:0]
d1[31:0]
sel
y[31:0]
Mux_instancia
u_alu
a[31:0]
alu_control[2:0]
b[31:0]
flag_z
result[31:0]
Top_ALU
u_pc_reg
clk
pc_next[31:0]
reset
pc[31:0]
program_counter
aluresult[31:0]
pc[31:0]
writedata[31:0]
zero
datapath
aluresult[31:0]
memwrite
pc[31:0]
writedata[31:0]
riscv_core
DataAdr[31:0]
MemWrite
WriteData[31:0]